Applications of Inner Product Spaces

- QR factorization
- Singular Value Decomposition
- Least square solutions

Learning objectives

- We will have a geometric understanding of the above topics which will lead to the definitions.
- Understand the concept of decomposing a matrix into an orthogonal matrix Q and an upper triangular matrix R.
- Interpret singular values in terms of matrix rank, range, and null space.
- Formulate least squares problems using matrix equations $Ax = B$.

1 QR factorization

In recent years, a numerical algorithm based on the Gram–Schmidt process, and known as QR**-decomposition**, has assumed growing importance as the mathematical foundation for a wide variety of numerical algorithms, including those for computing eigenvalues of large matrices.

The technical aspects of such algorithms are discussed here that specialize in the numerical aspects of linear algebra. However, we will discuss some of the underlying ideas here. We begin by posing the following concept.

QR decomposition of a matrix

Definition 1. If A is an $m \times n$ matrix with *linearly independent columns*, then A can be factored as

$$A = QR$$

where Q is an $m \times n$ matrix with orthonormal column vectors, and R is an $n \times n$ invertible upper triangular matrix.

To find this factorization, consider the following $m \times n$ matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Suppose that the column vectors of A are denoted by

$$u_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \quad u_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}, \quad \cdots, \quad u_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

Since $\{u_1, u_2, \cdots, u_n\}$ are linearly independent vectors in $\mathbb{R}^m$, we convert them into an **orthonormal** set of vectors $\{q_1, q_2, \cdots, q_n\}$ by the Gram Schmidt process (discussed in Module 4). The matrix Q is formed with the vectors $q_i's$ as the columns

$$Q = \begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{21} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{m1} & q_{m2} & \cdots & q_{mn} \end{pmatrix}$$

The matrix R can be calculated as $R = Q^T A$ or more explicitly

$$R = \begin{pmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \cdots & \langle u_n, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \cdots & \langle u_n, q_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \langle u_n, q_n \rangle \end{pmatrix}$$

Example 1. Find a QR-decomposition of the given 3×3 matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution The column vectors of A are

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

First, applying the Gram–Schmidt process with normalization gives

Step 1.

$$\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$$

Step 2.

$$\mathbf{v}_2 = \mathbf{u}_2 - \operatorname{proj}_{\mathbf{v}_1} \mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$$

$$= (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

Step 3.

$$\mathbf{v}_3 = \mathbf{u}_3 - \operatorname{proj}_{\mathbf{v}_1} \mathbf{u}_3 - \operatorname{proj}_{\mathbf{v}_2} \mathbf{u}_3$$

$$= \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$$

$$= (0, 0, 1) - \frac{1}{3}(1, 1, 1) - \frac{-\frac{2}{3}}{\frac{2}{3}} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

$$= \left(0, -\frac{1}{2}, \frac{1}{2} \right)$$

Thus,

$$\mathbf{v}_1 = (1, 1, 1), \qquad \mathbf{v}_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right), \qquad \mathbf{v}_3 = \left(0, -\frac{1}{2}, \frac{1}{2} \right)$$

form an orthogonal basis for $\mathbb{R}^3$. The norms of these vectors are

$$\|\mathbf{v}_1\| = \sqrt{3}, \qquad \|\mathbf{v}_2\| = \frac{\sqrt{6}}{3}, \qquad \|\mathbf{v}_3\| = \frac{1}{\sqrt{2}}.$$

Thus the corresponding orthonormal vectors are

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right),$$

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right),$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

$$\mathbf{q}_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Thus, Q is

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix},$$

Now, it follows from the above formula that we calculate R as

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \langle \mathbf{u}_3, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \langle \mathbf{u}_3, \mathbf{q}_2 \rangle \\ 0 & 0 & \langle \mathbf{u}_3, \mathbf{q}_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Thus, R is

$$R = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Hence, the QR-decomposition of A is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Example 2. Find the QR decomposition (or factorization) of matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

Solution Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of matrix A.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Use the Gram-Schmidt process to generate orthonormal vectors:

$$\mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} 0 \\ \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}.$$

We now write matrix Q whose columns are the vectors $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ 0 & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}.$$

We now calculate R:

$$R = Q^T A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & \frac{2}{\sqrt{5}} \end{bmatrix}.$$

We now write the given matrix A in factored form as follows:

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ 0 & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & \frac{2}{\sqrt{5}} \end{pmatrix}.$$

Example 3. Find the QR decomposition (or factorization) of matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Solution Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of matrix A:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Use the Gram-Schmidt process to generate orthonormal vectors:

$$\mathbf{q}_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

We now write matrix Q whose columns are the vectors $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$:

$$Q = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

We now calculate R:

$$R = Q^T A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & -\frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

We now write the given matrix A in factored form as follows:

$$A = QR = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & -\frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Example 4. Find the QR decomposition of the following 4×3 matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Solution Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of matrix A:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Use the Gram-Schmidt process to generate orthonormal vectors:

$$\mathbf{q}_1 = \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} -3/\sqrt{12} \\ 1/\sqrt{12} \\ 1/\sqrt{12} \\ 1/\sqrt{12} \end{pmatrix}, \quad \mathbf{q}_3 = \begin{pmatrix} 0 \\ -2/\sqrt{6} \\ 1\sqrt{6} \\ 1/\sqrt{6, m} \end{pmatrix}.$$

We now write matrix Q whose columns are the vectors $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$:

$$Q = \begin{pmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{pmatrix}$$

We now calculate R:

$$R = Q^T A = \begin{pmatrix} 2 & 3/2 & 1 \\ 0 & 3/\sqrt{12} & 2/\sqrt{12} \\ 0 & 0 & 2/\sqrt{6} \end{pmatrix}.$$

We now write the given matrix A in factored form as follows:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}_{4\times 3} = \begin{pmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{pmatrix}_{4\times 3} \begin{pmatrix} 2 & 3/2 & 1 \\ 0 & 3/\sqrt{12} & 2/\sqrt{12} \\ 0 & 0 & 2/\sqrt{6} \end{pmatrix}_{3\times 3}$$

2 Exercise

Find the QR-factorization of each matrix.

1. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 4 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$

6. $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix}$

Answers: QR-Factorizations

1. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{6}}{2} \end{bmatrix}$

$A = QR$

2. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$

$Q = \begin{bmatrix} \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{5}} \\ 0 & 0 \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{6}} & \frac{2}{\sqrt{5}} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{6} & \frac{5}{\sqrt{6}} \\ 0 & \sqrt{5} \end{bmatrix}$

$A = QR$

3. $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$Q = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad R = \begin{bmatrix} 2 & 2 & -\frac{1}{2} \\ 0 & 2 & \frac{1}{2} \\ 0 & 0 & \sqrt{2} \end{bmatrix}$

$A = QR$

4. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 4 \end{bmatrix} = QR$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \qquad R = \begin{bmatrix} \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$$

5. $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = QR$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}, \qquad R = \begin{bmatrix} \sqrt{5} & \sqrt{5} \\ 0 & \sqrt{5} \end{bmatrix}$$

6. $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix} = QR$

$$Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{38}}{38} & -\frac{3\sqrt{19}}{19} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{38}}{38} & \frac{3\sqrt{19}}{19} \\ 0 & \frac{3\sqrt{38}}{19} & \frac{\sqrt{19}}{19} \end{bmatrix}, \qquad R = \begin{bmatrix} \sqrt{2} & \frac{3\sqrt{2}}{2} & \sqrt{2} \\ 0 & \frac{\sqrt{38}}{2} & \frac{3\sqrt{38}}{19} \\ 0 & 0 & \frac{\sqrt{19}}{19} \end{bmatrix}$$

3 Singular Value Decomposition (SVD)

There are various ways to write a given matrix as a product of simpler matrices with special properties. For example, with the LU factorization an $m \times n$ matrix A could be written as $A = LU$ with L being an invertible lower triangular matrix and U an upper triangular matrix. We also saw the LDU decomposition of a matrix A where D is a diagonal matrix. In this section we consider a decomposition for $m \times n$ matrices. Specifically, we introduce the singular value decomposition, abbreviated as SVD, which enables us to write any $m \times n$ matrix as

$$A_{m\times n} = U_{m\times m}\Sigma_{m\times n}V^T_{n\times n}$$

where U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix and Σ is an $m \times n$ *diagonal* matrix with entries which are called the **singular values** of A.

The results that we will develop in this section have applications to compression, storage, and transmission of digitized information and forms the basis for many of the best computational algorithms that are currently available for solving linear systems.

Singular values

To define the singular values of an $m \times n$ matrix A, we consider the matrix A^TA. Observe that since A is an $m \times n$ matrix, the product A^TA is a square $n \times n$ matrix. This new matrix is also symmetric since $(A^TA)^T = A^T(A^T)^T = A^TA$. In such a case, there is a result that the eigenvalues λ_i of A^TA are all positive. This permits us to make the following definition.

Definition of singular values

Definition 2.
Let A be an $m \times n$ matrix. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A^TA, then the numbers

$$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}, \quad \cdots, \quad \sigma_n = \sqrt{\lambda_n}$$

are called the **singular values** of A.

Note 1. Since the eigenvalues λ_i of A^TA are positive, we can order them as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Hence we order the singular values as$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$

Singular Value Decomposition (Expanded Form)

We outline the steps to find the singular value decomposition of any $m \times n$ matrix A with $m \geq n$. Our goal is to find the matrices U, Σ, V such that

$$A = U\Sigma V^T$$

where U, Σ, and V have sizes $m \times m$, $m \times n$, and $n \times n$, respectively.

- First find the eigenvalues of the square symmetric matrix A^TA. Order them as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Find the singular values $\sigma_i = \sqrt{\lambda_i}$, $i = 1, 2, \cdots, n$ and order them as

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n.$$

Note that this order is important and the applications of SVD assumes that the singular values are ordered in the decreasing order.

- Find the eigenvectors of the matrix $A^T A$. Let the eigenvector be denoted as v_i corresponding to the eigenvalue λ_i. Write the eigenvectors as the (ordered) columns to get the matrix V. Thus if the eigenvectors are

$$v_1 = \begin{pmatrix} v_{11} \\ v_{21} \\ \vdots \\ v_{n1} \end{pmatrix}, \quad v_2 = \begin{pmatrix} v_{12} \\ v_{22} \\ \vdots \\ v_{n2} \end{pmatrix}, \quad \cdots \quad , v_n = \begin{pmatrix} v_{1n} \\ v_{2n} \\ \vdots \\ v_{nn} \end{pmatrix}$$

corresponding to the eigenvalues λ_1, λ_2, $\cdots$, λ_n respectively, then the matrix V is

$$V = \begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{pmatrix}_{n \times n}$$

- Find the vectors

$$u_i = \frac{1}{\sigma_i} A v_i, \quad i = 1, 2, \cdots, n.$$

Note that U is an $m \times m$ matrix with $m \geq n$. The n vectors u_i form the first n columns of U. The remaining $m - n$ columns are added to U so that U remains an orthogonal matrix. The method to add these $m - n$ columns is explained in the examples that follow. It involves finding a basis for the solution space of a homogeneous system of equation followed by Gram-Schmidt method to convert this basis into an orthogonal basis.

- The *diagonal* matrix Σ is of the same order $m \times n$ as the given matrix A. The matrix Σ is

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \sigma_n & \cdots & 0 & 0 \\ 0 & 0 & 0 & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{pmatrix}$$

That is, in the matrix Σ, the n singular values σ_i are first written as diagonal elements and then the rest of the entries are filled with zeros to make the order of Σ as $m \times n$.

Example 5. Find a singular value decomposition of the matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

Solution The first step is to find the eigenvalues of the matrix

$$A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of $A^T A$ is

$$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

so the eigenvalues of $A^T A$ are

$$\lambda_1 = 3 \qquad \text{and} \qquad \lambda_2 = 1,$$

and the singular values of A in order of decreasing size are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}, \qquad \sigma_2 = \sqrt{\lambda_2} = 1.$$

We leave it for you to verify that

$$\mathbf{v}_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \qquad \text{and} \qquad \mathbf{v}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}$$

are eigenvectors corresponding to λ_1 and λ_2, respectively, and that

$$V = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix}$$

orthogonally diagonalizes $A^T A$.
From part (d) of above Theorem, the vectors

$$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1 = \frac{\sqrt{3}}{3} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}$$

and

$$\mathbf{u}_2 = \frac{1}{\sigma_2} A\mathbf{v}_2 = (1) \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

are two of the three column vectors of U. Note that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthonormal, as expected. We could extend the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ to an orthonormal basis for $\mathbb{R}^3$. However,

the computations will be easier if we first remove the messy radicals by multiplying $\mathbf{u}_1$ and $\mathbf{u}_2$ by appropriate scalars. Thus, we will look for vector $\mathbf{u}_3$ that is orthogonal to

$$\sqrt{6}\,\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \qquad \text{and} \qquad \sqrt{2}\,\mathbf{u}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

To satisfy these two orthogonality conditions, the vector $\mathbf{u}_3$ must be a solution of the homogeneous linear system

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We leave it for you to show that a general solution of this system is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Normalizing the vector on the right yields

$$\mathbf{u}_3 = \begin{bmatrix} -\dfrac{1}{\sqrt{3}} \\ \dfrac{1}{\sqrt{3}} \\ \dfrac{1}{\sqrt{3}} \end{bmatrix}$$

Thus, the singular value decomposition of A is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \dfrac{\sqrt{6}}{3} & 0 & -\dfrac{1}{\sqrt{3}} \\ \dfrac{\sqrt{6}}{6} & -\dfrac{\sqrt{2}}{2} & \dfrac{1}{\sqrt{3}} \\ \dfrac{\sqrt{6}}{6} & \dfrac{\sqrt{2}}{2} & \dfrac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dfrac{\sqrt{2}}{2} & \dfrac{\sqrt{2}}{2} \\ \dfrac{\sqrt{2}}{2} & -\dfrac{\sqrt{2}}{2} \end{bmatrix}^T$$

$$A = U\Sigma V^T$$

You may want to confirm the validity of this equation by multiplying out the matrices on the right side.

Example 6. Find a singular value decomposition of the matrix $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$.

Solution

Step 1: Compute A^TA

$$A^TA = \begin{bmatrix} 1 & 3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}.$$

Step 2: Eigenvalues and eigenvectors

Eigenvalues:

$$\lambda_1 = 18, \qquad \lambda_2 = 2.$$

Eigenvectors:

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus,

$$V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Step 3: Singular values

$$\sigma_1 = \sqrt{18} = 3\sqrt{2}, \qquad \sigma_2 = \sqrt{2}.$$

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}.$$

Step 4: Compute AA^T

$$AA^T = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 18 \end{bmatrix}.$$

Eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Thus,

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Final SVD

$$A = U\Sigma V^T.$$

Example 7. Find a singular value decomposition of the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}.$$

Solution First, compute

$$A^T A = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}.$$

The eigenvalues of A^TA are 18 and 0, with corresponding unit eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

These unit vectors form the columns of V:

$$V = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

The singular values are

$$\sigma_1 = \sqrt{18} = 3\sqrt{2} \qquad \text{and} \qquad \sigma_2 = 0.$$

Since there is only one nonzero singular value, the "matrix" D may be written as a single number. That is,

$$D = 3\sqrt{2}.$$

The matrix Σ is the same size as A, with D in its upper left corner:

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

To construct U, first construct $A\mathbf{v}_1$ and $A\mathbf{v}_2$:

$$A\mathbf{v}_1 = \begin{bmatrix} \frac{2}{\sqrt{2}} \\ -\frac{4}{\sqrt{2}} \\ \frac{4}{\sqrt{2}} \end{bmatrix}, \qquad A\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

As a check on the calculations, verify that

$$\|A\mathbf{v}_1\| = \sigma_1 = 3\sqrt{2}.$$

Of course,

$$A\mathbf{v}_2 = 0$$

because

$$\|A\mathbf{v}_2\| = \sigma_2 = 0.$$

The only column found for U so far is

$$\mathbf{u}_1 = \frac{1}{3\sqrt{2}} A\mathbf{v}_1 = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{bmatrix}.$$

The other columns of U are found by extending the set $\{\mathbf{u}_1\}$ to an orthonormal basis for $\mathbb{R}^3$. In this case, we need two orthogonal unit vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ that are orthogonal to $\mathbf{u}_1$. Each vector must satisfy

$$\mathbf{u}_1^T \mathbf{x} = 0,$$

which is equivalent to the equation

$$x_1 - 2x_2 + 2x_3 = 0.$$

A basis for the solution set of this equation is

$$\mathbf{w}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{w}_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

(Check that $\mathbf{w}_1$ and $\mathbf{w}_2$ are each orthogonal to $\mathbf{u}_1$.)
Apply the Gram–Schmidt process (with normalizations) to $\{\mathbf{w}_1, \mathbf{w}_2\}$, and obtain

$$\mathbf{u}_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}, \qquad \mathbf{u}_3 = \begin{bmatrix} -\frac{2}{\sqrt{45}} \\ \frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{bmatrix}.$$

Finally, set

$$U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix},$$

take Σ and V^T from above, and write

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & -\frac{2}{\sqrt{45}} \\ -\frac{2}{3} & \frac{1}{\sqrt{5}} & \frac{4}{\sqrt{45}} \\ \frac{2}{3} & 0 & \frac{5}{\sqrt{45}} \end{bmatrix} \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Exercises

Find the SVD of each matrix.

1. $\begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix}$

2. $\begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$

3. $\begin{bmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{bmatrix}$

4. $\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$

Answers

1. $A = \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T$

   $A = U\Sigma V^T$

2. $A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$

   $U = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \qquad V = \begin{bmatrix} -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$

   $A = U\Sigma V^T$

3. $A = \begin{bmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{bmatrix}$

   $U = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 3\sqrt{10} \\ 0 & 0 \end{bmatrix}, \qquad V = \begin{bmatrix} -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$

   $A = U\Sigma V^T$

4. $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$

$$U = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 3\sqrt{10} \\ 0 \\ 0 \end{bmatrix}, \qquad V = \begin{bmatrix} -\frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}$$

$$A = U\Sigma V^T$$

4 Least square solutions

Suppose that $Ax = b$ is an **inconsistent** linear system of m equations in n unknowns (so that A is an $m \times n$ matrix, x is an $n \times 1$ vector and b is an $m \times 1$ vector) in which we suspect the inconsistency to be caused by errors in the entries of A or b. Since no exact solution is possible, that is there is no $x \in \mathbb{R}^n$ such that Ax is exactly equal to b, the best one can do is to find a vector $\hat{x} \in \mathbb{R}^n$ that makes $A\hat{x}$ as close as possible to b.

If we think of Ax as an approximation to b, the **least square problem** is to find an $\hat{x} \in \mathbb{R}^n$ that makes the distance between b and Ax as small as possible, that is the distance between b and $A\hat{x}$ is the smallest among all vectors $x \in \mathbb{R}^n$. This $\hat{x} \in \mathbb{R}^n$ is called the **least square solution** of the system of equations $Ax = b$.

It is important to note that in an inconsistent system of equations is that for every x in $\mathbb{R}^n$, the vector Ax lies in the column space of A while b does not lie in the column space of A (refer to the section on column space in Module 2).

However, we can project b onto the column space of A (orthogonal projection method of Module 4). Let

$$\hat{b} = \text{proj}_{ColA} b$$

Since $\hat{b}$ lies in the column space of A, the system $Ax = \hat{b}$ is consistent, and there is a vector $\hat{x} \in \mathbb{R}^n$ which is a solution, that is, $A\hat{x} = \hat{b}$.

This is illustrated in Figure (a), while in Figure (b) we have that $A\hat{\mathbf{x}}$ is the orthogonal projection of b on the column space of A, that is,

$$A\hat{\mathbf{x}} = \text{proj}_{\text{col}(A)}\, \mathbf{b}.$$



(a) **(b)**

Note that we have actually solved the least square problem is we use the following result.

Theorem 1. Let W be a subspace of $\mathbb{R}^n$, y any vector in $\mathbb{R}^n$ and $\hat{y}$ the orthogonal projection of y onto W. Then $\hat{y}$ is the closest point in W to y, in the sense that the distance between y and $\hat{y}$ is the least compared to the distance between y and any other vector in W.

In the present context, the subspace W is the column space of A and the vectors y and $\hat{y}$ are the vectors b and $\hat{b}$ respectively. Thus, by the above result, $\hat{b} = A\hat{x}$ is the closest point in the column space of A that is closest to b so that the distance between $A\hat{x}$ and b is the least compared to the distance between b and any other vectors in the column space of A. The least square solution is the solution $\hat{x}$ of the system of equation $Aw = \hat{b}$. Thus we have the following working rule

- Given the inconsistent system of equations $Ax = b$, find the orthogonal projection $\hat{b}$ of b onto the column space of A (orthogonal decomposition theorem, Module 4).
- The system $Aw = \hat{b}$ always has a solution. A solution $\hat{x}$ of $Aw = \hat{b}$ is the least square solution of the system $Ax = b$.

What we have discussed so far is the general theory of the least square problem. Using the fact that $b - Ax$ is orthogonal to the column space of A (refer Figure (a)) and hence to all the columns of the matrix A, we have the following result

Theorem 2. The set of least-squares solutions of

$$A\mathbf{x} = \mathbf{b}$$

coincides with the nonempty set of solutions of the **normal equations**

$$A^T A\mathbf{x} = A^T\mathbf{b}.$$

The above result gives a direct working rule to solve the least square problem, which will be used to solve all the problems in the module.

- Given the inconsistent system of equations $Ax = b$, find $A^T A$ and $A^T b$.
- The normal equations $A^T Aw = A^T b$ always has a solution. A solution $\hat{x}$ of $A^T Aw = A^T b$ is a least square solution of the system $Ax = b$.

Note 2. Suppose $\hat{x}$ is found from the normal equations.

- From the first working rule, $A\hat{x}$ is the projection of b onto the column space of A and so $A\hat{x}$ is the vector that is closest to b in the column space of A.
- The **least square error** is the magnitude of the vector $A\hat{x} - b$ given by $||A\hat{x} - b||$. Hence the distance between b and any vector in the column space of A is at **least** $||A\hat{x} - b||$.

In case the columns of A are linearly independent, we have another method to compute the least square solution though it is computationally not efficient.

Theorem 3. Let A be an $m \times n$ matrix. The matrix $A^T A$ is invertible if and only if the columns of A are linearly independent. In this case the equation $Ax = b$ has a unique least square solution $\hat{x}$ given by

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b}.$$

Example 8. Find a least-squares solution of the inconsistent system

$$A\mathbf{x} = \mathbf{b}$$

for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}.$$

Solution To use normal equations, compute:

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

and

$$A^T \mathbf{b} = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}.$$

Then the equation

$$A^T A\mathbf{x} = A^T \mathbf{b}$$

becomes

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}.$$

Row operations can be used to solve this system, but since $A^T A$ is invertible and 2×2, it is probably faster to compute

$$(A^T A)^{-1} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

and then to solve

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$$

as

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

$$= \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Example 9. (a) Show that the system $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} \qquad \text{and} \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

is inconsistent (a system with no solutions).

(b) Find the least-squares solution to the inconsistent system $A\mathbf{x} = \mathbf{b}$.

(c) Determine the least-squares error given by

$$\|A\hat{\mathbf{x}} - \mathbf{b}\|$$

for the solution found in part (b).

Solution

(a)

The system of equations may be written in the form

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}.$$

Write the above system in augmented matrix form:

$$\left[\begin{array}{cc|c} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 3 \end{array}\right].$$

Row reduce (Gauss–Jordan) the augmented matrix:

$$\left[\begin{array}{cc|c} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 2 & \frac{5}{2} \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & \frac{5}{2} \end{array}\right].$$

The last row gives the equation

$$0x_1 + 0x_2 = \frac{5}{2},$$

which has no solution. Therefore, the system is inconsistent.

(b)

First compute

$$A^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Now calculate A^TA and $A^T\mathbf{b}$:

$$A^TA = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix},$$

$$A^T\mathbf{b} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Substituting into the normal equation

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b},$$

we obtain

$$\begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving the system gives

$$\hat{\mathbf{x}} = \begin{bmatrix} \frac{13}{21} \\ \frac{20}{21} \end{bmatrix}.$$

(c)

Compute the least-squares error:

$$A\hat{\mathbf{x}} - \mathbf{b} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{13}{21} \\ \frac{20}{21} \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{5}{21} \\ \frac{20}{21} \\ -\frac{10}{21} \end{bmatrix}.$$

Hence,

$$\|A\hat{\mathbf{x}} - \mathbf{b}\| = \sqrt{\left(\frac{5}{21}\right)^2 + \left(\frac{20}{21}\right)^2 + \left(-\frac{10}{21}\right)^2} = \sqrt{\frac{25}{441} + \frac{400}{441} + \frac{100}{441}} = \sqrt{\frac{525}{441}} = \frac{5\sqrt{21}}{21} \approx 1.1.$$

Exercises

Find the least square solution of $A\mathbf{x} = \mathbf{b}$. Also, compute the least square error in each case.

1. $A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$

6. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 7 \\ 2 \\ 3 \\ 6 \\ 5 \\ 4 \end{bmatrix}$

Answers

1. $\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

2. $\hat{\mathbf{x}} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$

3. $\hat{\mathbf{x}} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}$

4. $\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

5. $\hat{\mathbf{x}} = \begin{bmatrix} \frac{7}{3} \\ -\frac{1}{3} \\ \frac{8}{3} \end{bmatrix}$

6. $\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{5}{2} \end{bmatrix}$

5 Additional Exercises

1. Prove the following QR factorization of the given matrices

(a) $$\begin{pmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{pmatrix} = \frac{\sqrt{5}}{10} \begin{pmatrix} 3 & 1 & -3 \\ 1 & 3 & 1 \\ -1 & 3 & 1 \\ 3 & -1 & 3 \end{pmatrix} \begin{pmatrix} 2\sqrt{5} & -4\sqrt{5} & 3\sqrt{5} \\ 0 & 2\sqrt{5} & -\sqrt{5} \\ 0 & 0 & 2\sqrt{5} \end{pmatrix}$$

(b) $$\begin{pmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} -\sqrt{3} & 3\sqrt{3} & -\sqrt{3} \\ 3\sqrt{3} & \sqrt{3} & -\sqrt{3} \\ \sqrt{3} & \sqrt{3} & 3\sqrt{3} \\ \sqrt{3} & -\sqrt{3} & -\sqrt{3} \end{pmatrix} \begin{pmatrix} 2\sqrt{3} & -6\sqrt{3} & \sqrt{3} \\ 0 & 2\sqrt{3} & 5\sqrt{3} \\ 0 & 0 & 2\sqrt{3} \end{pmatrix}$$

(c) $$\begin{pmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & -1 \\ 1 & 5 \\ -3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 6 & 12 \\ 0 & 6 \end{pmatrix}$$

(d) $$\begin{pmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} -2 & 5 \\ 5 & 2 \\ 2 & -4 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 7 & 7 \\ 0 & 7 \end{pmatrix}$$

(e) $$\begin{pmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ 2 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 0 & 1 \end{pmatrix}$$

(f) $$\begin{pmatrix} 1 & -1 \\ 1 & 4 \\ 1 & -1 \\ 1 & 4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 0 & 5 \end{pmatrix}$$

(g) $$\begin{pmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{pmatrix} = \frac{1}{42} \begin{pmatrix} 14\sqrt{3} & -9\sqrt{14} & \sqrt{42} \\ 14\sqrt{3} & 3\sqrt{14} & -5\sqrt{42} \\ 14\sqrt{3} & 6\sqrt{14} & 4\sqrt{42} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 3\sqrt{3} & 0 \\ 0 & 2\sqrt{14} & \sqrt{14} \\ 0 & 0 & 0 \end{pmatrix}$$

(h) $\begin{pmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{pmatrix}$

2. Prove the following singular value decomposition of the given matrices

(a) $\begin{pmatrix} -1 & 1 \\ -1 & 1 \\ 2 & -2 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} -\sqrt{6} & -3\sqrt{2} & 2\sqrt{3} \\ -\sqrt{6} & 3\sqrt{2} & 2\sqrt{3} \\ 2\sqrt{6} & 0 & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} 2\sqrt{3} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} \sqrt{2} & \sqrt{2} \\ -\sqrt{2} & \sqrt{2} \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 1 \\ 3 & -3 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} -2 & -2 \\ 1 & 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -2\sqrt{5} & \sqrt{5} \\ \sqrt{5} & 2\sqrt{5} \end{pmatrix} \begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}$

(d) $\begin{pmatrix} -1 & -2 \\ 1 & -2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & -1 \\ -2 & 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2 & 0 \\ -\sqrt{2} & 0 & \sqrt{2} \\ -\sqrt{2} & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 2\sqrt{3} & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{30} \begin{pmatrix} -10\sqrt{6} & 6\sqrt{5} & -2\sqrt{30} \\ 5\sqrt{6} & 0 & -5\sqrt{30} \\ 5\sqrt{6} & 12\sqrt{5} & \sqrt{30} \end{pmatrix}$

(f) $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -\sqrt{2} & 0 & \sqrt{2} \\ 0 & 2 & 0 \\ \sqrt{2} & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} -\sqrt{2} & 0 & \sqrt{2} \\ \sqrt{2} & 0 & \sqrt{2} \\ 0 & 2 & 0 \end{pmatrix}$

(g) $\begin{pmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{pmatrix} = \frac{1}{15} \begin{pmatrix} -5 & 6\sqrt{5} & 2\sqrt{15} \\ 10 & 3\sqrt{5} & -4\sqrt{5} \\ 10 & 0 & 5\sqrt{5} \end{pmatrix} \begin{pmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{10} \begin{pmatrix} 3\sqrt{10} & \sqrt{10} \\ -\sqrt{10} & 3\sqrt{10} \end{pmatrix}$

(h) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & -3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 0 & -2\sqrt{6} \\ 2\sqrt{3} & 3\sqrt{2} & \sqrt{6} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$

3. Find the least square solutions (given as $\hat{x}$) of $Ax = b$, given that

(a) $A = \begin{pmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 7/6 \\ 1 \end{pmatrix} \right)$

(b) $A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} -5 \\ 8 \\ 1 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} -7/6 \\ 11/15 \end{pmatrix} \right)$

(c) $A = \begin{pmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ 1 \\ -4 \\ 2 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix} \right)$

(d) $A = \begin{pmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$

(e) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 3 \\ 8 \\ 2 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} t \\ 2-t \\ 5-t \end{pmatrix}, \quad t \in \mathbb{R} \right)$

4. In the following exercises, find (a) the least square solution (given as $\hat{x}$) of $Ax = b$ and (b) the orthogonal projection of b onto the column space of A (given as P).

(a) $A = \begin{pmatrix} 1 & 5 \\ 3 & 1 \\ -2 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ -2 \\ -3 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 2/7 \\ 1/7 \end{pmatrix}, \quad P = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right)$

(b) $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 3 \\ 1/2 \end{pmatrix}, \quad P = \begin{pmatrix} 4 \\ -1 \\ 4 \end{pmatrix} \right)$

(c) $A = \begin{pmatrix} 4 & 0 & 1 \\ 1 & -5 & 1 \\ 6 & 1 & 0 \\ 1 & -1 & -5 \end{pmatrix}, \quad b = \begin{pmatrix} 9 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 2/3 \\ 0 \\ 1/3 \end{pmatrix}, \quad P = \begin{pmatrix} 3 \\ 1 \\ 4 \\ -1 \end{pmatrix} \right)$

(d) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 5 \\ 6 \\ 6 \end{pmatrix}, \qquad \left(\hat{x} = \begin{pmatrix} 1/3 \\ 14/3 \\ 5/3 \end{pmatrix}, \quad P = \begin{pmatrix} 5 \\ -4/3 \\ 19/3 \\ 8/3 \end{pmatrix} \right)$

5. Find the least square error associated with all the least-square problems given in $Q4$.

(a) $e = 27$

(b) $e = 2$

(c) $e = 54$

(d) $e = 181/3$